What You Can Cram Into a Single Vector: Probing Sentence Embeddings (https://arxiv.org/pdf/1805.01070)

Conneau et al. (2018) test what information survives when a sentence is squeezed into a single embedding vector. They freeze the embedding and train a small classifier, called a probe, to predict one simple property of the original sentence from the vector alone. If the probe gets it right, that property is stored in the vector. If it can't do better than chance, the information was lost or is buried too deep to read. They build 10 of these tests, ranging from surface features (sentence length, which words appear) to grammar (word order, parse tree depth) to meaning (verb tense, whether a word was swapped for one that doesn't fit). They run all 10 tests on many encoders and compare the results.

E.g. To check whether an embedding remembers verb tense, they embed thousands of sentences labeled "past" or "present" and train a probe to predict the label from the vector. The probe reaches about 90% accuracy, so tense is clearly stored. On the word order test, which asks whether two neighboring words were swapped, a simple average of word vectors scores 50%, the same as guessing. That shows averaging throws word order away completely. They also find that some information fades as a model trains: sentence length becomes harder to read out over time, so models actively drop details they don't need.